

Question 1 – ER diagram for a Healthcare Clinic System

Case: Designing a Database for a Healthcare Clinic

During interviews with clinic administrators and medical staff, the following requirements were gathered to design a robust and scalable database system for managing patients, doctors, medical services, and equipment reservations. The goal is to improve patient care, resource allocation, and service tracking within the clinic.

Primary Requirements:

The clinic needs to manage doctors related information, patient records, appointments, and healthcare services.

Each doctor can treat multiple patients and provide multiple services.

The system should track appointments, including the involved doctor, patient, and the services performed.

The clinic also manages medical devices that can be reserved and assigned to appointments.

Detailed Requirements:

Doctor Characteristics:

Each doctor has a unique employee ID, full name, specialization (e.g., cardiology, dermatology), and contact info.

Doctors can provide multiple types of services.

Patient Characteristics:

Each patient has a unique health ID, name, date of birth, and contact info.

Patients can book appointments with multiple doctors and receive multiple services over time.

The database must track the appointment date, time, and visit reason.

Medical Service Characteristics:

Each service has a unique code, name (e.g., "Blood Test", "ECG"), and description.

Services may have required equipment/devices associated with them.

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Equipment Characteristics:

Equipment includes items such as ultrasound machines, ECG monitors, and portable scanners.

Each device has a serial number, type, availability status (available, reserved, under maintenance), and model.

Devices can be reserved for specific appointments.

Appointment Characteristics:

An appointment links a patient, a doctor, and one or more services.

Each appointment has a scheduled date/time, status (scheduled, completed, cancelled), and optional notes.

Equipment reservations must be tied to appointments where relevant.

Using draw.io, design an Entity-Relationship diagram that models the full scope of this healthcare clinic system.

Your ERD should:

Represent all entities and their attributes clearly.

Include all relevant relationships and constraints (e.g., one-to-many, many-to-many).

Show cardinality, participation, and optionality as appropriate.

In addition to the ER diagram, submit a one-page table that explicitly maps the given written requirements to elements in your ER diagram. This table must include the following columns:

- Requirement statement (copied from the assignment)
- Corresponding entity or relationship in the ERD
- Cardinality and participation decision
- Brief justification (2–3 sentences)

Moreover, include a short-written section (maximum one page) describing:

- Three assumptions your group made when interpreting the requirements
- Two alternative design choices that were considered but not selected, along with a brief explanation of why they were rejected

Submission Instructions:

- Submit a PDF file showing your ER diagram by the due date of this assignment.

Question 2 SQL Queries

The question evaluates students' ability to understand and code SQL queries specially the followings:

1. apply simple queries using the SELECT statement.

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1. implement filtering with the WHERE statement.

employ arithmetic and string functions. Write a query using MySQL to return all countries included in the World database.

Write a query using MySQL that returns all countries in Europe.

Write a query using MySQL to return all countries that speaks English as an official language with percentage greater than or equal to 70.

Write a query using MySQL to return all countries that received their independence in the 18th century.

Write a query using MySQL that returns the following dataset.

This question concerns the <i>World</i>	CountryCode	District
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database that is installed with MySQL. Name		
Delhi	IND	Delhi
Ludhiana	IND	Punjab
Amritsar	IND	Punjab
Jalandhar (Jullundur)	IND	Punjab
New Delhi	IND	Delhi
Patiala	IND	Punjab
Bhatinda (Bathinda)	IND	Punjab
Pathankot	IND	Punjab
Hoshiarpur	IND	Punjab
Moga	IND	Punjab
Abohar	IND	Punjab
Delhi Cantonment	IND	Delhi

Instruction:

For each SQL query, include a structured SQL comment immediately above the query explaining:

- The purpose of the query
- The table(s) involved
- The filtering logic used
- Why does the query output satisfy the requirement

In addition to the SQL file, submit a single PDF containing screenshots of each query executed in MySQL Workbench. Each screenshot must clearly show:

- The SQL query
- The query result output
- The MySQL Workbench interface
- A visible timestamp